

Advanced Empirical Methods **Research Project**

Stephanie Pacheco
April 29, 2026

Introduction

Research Question: How does limited English proficiency influence household income among immigrants in the United States?

Null Hypothesis: Limited English proficiency has no effect on median household income among immigrants residing in the United States.

Alternative Hypothesis: Limited English proficiency leads to a decrease in median household income among immigrants residing in the United States.

Causal Mechanism:

- **Human Capital Theory:** Argues that an individual's skills and education are imperative to upward mobility in the labor market, with proficiency in the receiving country's language being a critical determinant to economic success → English language as a form of human capital
- **Structural barriers:** Limited English proficiency as a notable marker of foreignness → potential workplace discrimination and social isolation
- **Asset:** English proficiency facilitates job search and information accessibility → job stability



Data & Methods

Data Source: 2022 American Community Survey

Operationalization:

- **Immigrant population:** Total number of respondents who reported a place of birth outside the United States
- **Independent Variable** → Limited English proficiency: foreign-born who reported primarily speaking other languages and English less than “very well”
 - **Primary IV** → *Percentage* of Immigrants with limited English proficiency to make values comparable and meaningful measures across states
- **Dependent Variable** → Household Income: median household income (USD) in the past year; midpoint of the income distribution by state
- **Controls:** Age, Years of Entry, Education, Cost of Living, Unemployment Rate

Limitations

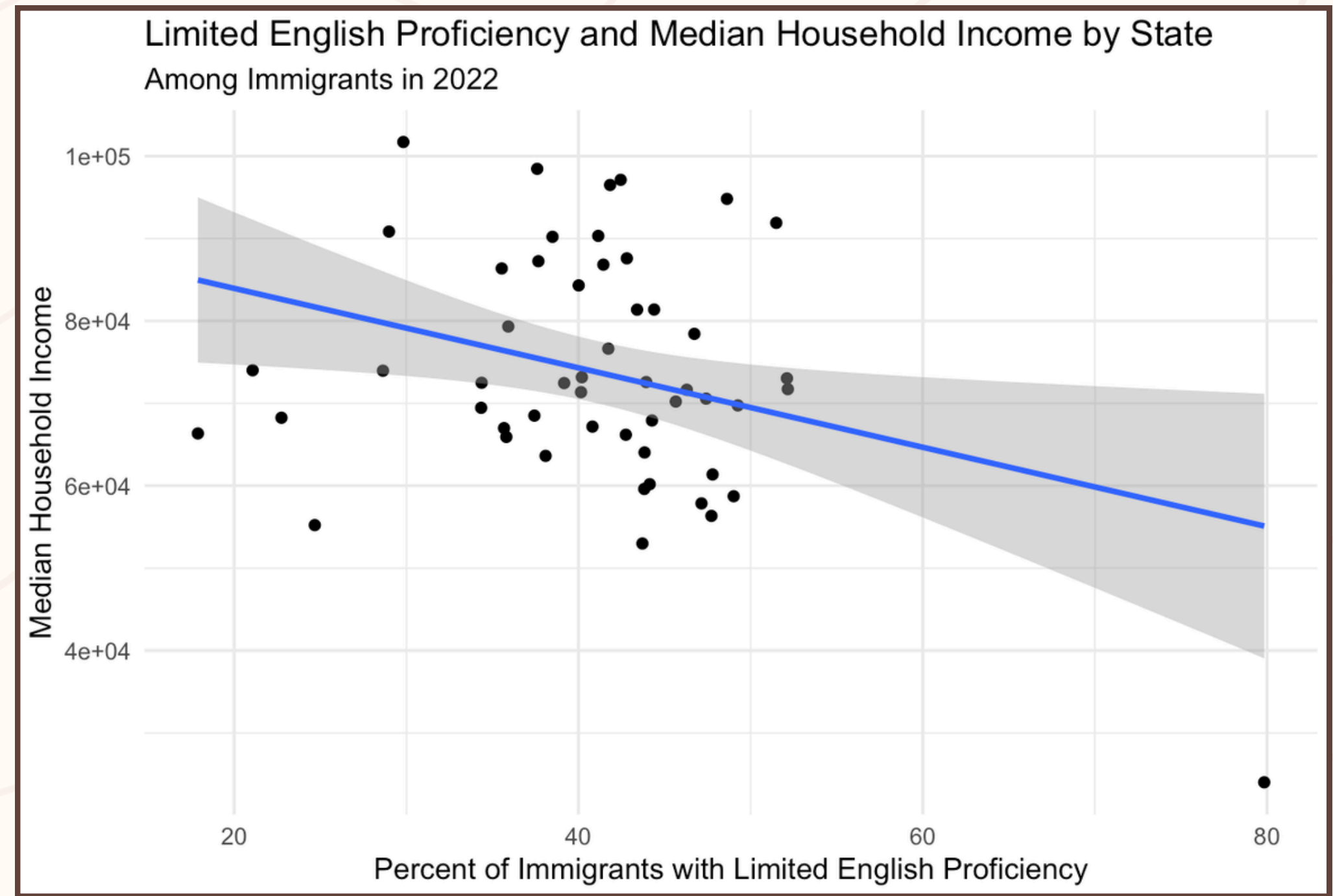
- Self-reported responses → subjective reporting
- Total household income reflects earnings from multiple household members → challenging to isolate the effect of English proficiency
- Aggregated variables at the state level → limits the ability to analyze individual experiences

Descriptives & Scatterplot

Table 1: Descriptive Statistics of Analytic Sample (By State in 2022)

Variable	Mean	Median	Standard Deviation	Min	Max
Total Foreign-born Population	872,510.2	265,770	1,745,496	19,756	10,442,880
Immigrants with Limited English Proficiency	398,574.7	115,041.5	874,971.4	4,422	5.376,985
Percentage of Immigrants with Limited English Proficiency (%)	41	41.78	9.49	17.89	79.83
Median Household Income (\$)	73,829	72,090	14,182.8	24,002	101,722

Source: American Community Survey, 2022 (ACS)



Preliminary Findings

Analytical Approach and Reasoning: Linear Regression model → analysis seeks to examine a linear relationship between the two continuous variables

Preliminary Findings: The baseline model suggests that for every 1% point increase in immigrants with limited English proficiency, the median household income in the state decreases by about \$482. This result is statistically significant at the 0.05 level.

Table 2: Linear Regression of Median Household Income on Immigrants with Limited English Proficiency (By State in 2022)		
	Model 1 Coefficient	Standard Error
Percentage of Immigrants with Limited English Proficiency	-482.2*	200.0
+ p < 0.1, * p < 0.05, ** p < 0.01 , *** p < 0.001		
Source: American Community Survey, 2022 (ACS)		

Discussion

Interpretation: The preliminary findings support the theoretical explanation and alternative hypothesis that higher levels of limited English proficiency among immigrants are associated with lower median household income across states.

- Suggests: limited English proficiency reduces access to opportunities for upward mobility and limits the ability to effectively navigate labor markets

Implications:

- Expanding the number of and access to English language learning programs nationwide
- Increasing multilingual resources to reduce language barriers

Thank **You!**
